

Story Embeddings - Narrative-Focused Representations of Fictional Stories

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Abstract

We present a novel approach to modeling fictional narratives. The proposed model creates embeddings that represent a story such that similar narratives, that is, reformulations of the same story, will result in similar embeddings. We showcase the prowess of our narrative-focused embeddings on various datasets, exhibiting state-of-the-art performance on multiple retrieval tasks. The embeddings also show promising results on a narrative understanding task. Additionally, we perform an annotation-based evaluation to validate that our introduced computational notion of narrative similarity aligns with human perception. The approach can help to explore vast datasets of stories, with potential applications in recommender systems and in the computational analysis of literature.

1 Introduction

Narrative understanding is a field that has received much attention in the last few years. Various approaches have tested models either on narrative-based question answering tasks or performed intrinsic evaluations, such as narrative cloze evaluations, where models need to predict missing events in a sequence.

In this work, we seek to address the topic of story embeddings with a focus on narrative, meaning representations that prioritize the aspect of ‘what’ is happening rather than the surface-level information of ‘how’ it is being told. For example, a love story with a specific twist can be set in different settings (outer space or countryside), with a different cast (e.g., different names and some different traits for all characters), or in a shortened version, without fundamentally changing the narrative. After altering the story’s final twist, the new narrative could still be considered similar without being identical.

We assume that any fictional text can be represented by its summary for our purposes of modeling the narrative. While there are various characteristics of a story that can not be gleaned from a summary, such as the style and mood of a text, the narrative is core to what is represented in a summary. Thus, summaries are the perfect testing ground for narrative embeddings, although an expansion to full texts in the future is desirable.

It has been observed that retellings of – specifically fairytales – have recently increasingly been published, with many retellings changing the setting to a modern-day one or introducing the representation of minorities (Goldman, 2023). As such, they represent a structurally similar story, with a new setting and limited alterations to the narrative. Other retellings, however, change the story significantly, sometimes merely retaining themes from the original work. On a limited scale, previous work has addressed the automatic identification of stories following the same plot (Glass, 2022). In this work, we consider this task as a possible application of story embeddings.

Researchers in the ACL community have, in the context of fictional works, often used the terms narrative and story without a clear distinction (e.g., [MISSING])

2 Related Work

A substantial line of work (e.g. Chambers and Jurafsky, 2008, 2009; Granroth-Wilding and Clark, 2016) has dealt with graph-based representations of narratives, specifically with predicting missing narrative triples and inferring schemas of commonly re-occurring narratives. Lee and Jung (2020) take what can be considered a hybrid approach, building explicit networks but using contextual vector representations rather than lexical items to represent triples. Similarly, using less contextual information, in prior work, we trained narrative triple embeddings based on narrative chains (Hatzel and Biemann, 2023). Following ever-increasing advancements in the field of language models and motivated by the information loss inherent to extracting narrative triples, this work seeks to apply a more distantly supervised approach to representing stories.

Our work builds on two previously released datasets (Hatzel and Biemani, 2024; Chaturvedi et al., 2018). Both datasets contain story summaries extracted from Wikipedia. Specifically, both seek to find different formulations of summaries for very similar stories. The movie remake dataset by Chaturvedi et al. (2018) contains a relatively small collection of summaries from multiple remakes of the same movie. In contrast, our previously

released dataset, Tell-Me-Again (Hatzel and Biemann, 2024), collects summaries from multiple Wikipedia language versions of the same fictional work. The movie remake dataset only contains 266 summaries and is thus not suited for training, whereas Tell-Me-Again contains roughly 30,000 stories. Each story comes with up to five different summaries, originally extracted from multiple Wikipedia language versions and automatically translated into English. The dataset additionally comes with a pseudonymized variant, explicitly created for training models that do not focus on entity names’. In this variant, entity names are replaced in each summary by alternatives in an internally consistent manner. These pseudonymized versions are created using rule-based replacement strategies on top of a model-based coreference resolution system’.

ROCStories is a dataset for testing commonsense reasoning, first released in 2016 (Mostafazadeh et al., 2016) with the introduction of the Story Cloze Task. In the task, systems pick one of two endings of a five-sentence story. One choice is a logical [ILLEGIBLE] conclusion to the story. As a result, humans can solve the Story Cloze Task perfectly, but at the time of publication, the best performing system in an accompanying shared task reached only around 75

The creation of semantic sentence representations with large language models (LLMs) has recently gained much interest. While Wang et al. (2024) train embeddings from last token hidden states, it has been suggested that the causal attention mechanism in only models limits their effectiveness for embeddings (Behnam Ghader et al., 2024). Alternatives have been based approaches to document embeddings. While it was primarily evaluated on short documents, it does not have based approaches.

The definition of what exactly constitutes narrative similarity has been addressed by Chen et al. (2022a) in their corresponding codebook (Chen et al., 2022b). In a pairwise similarity annotation task, they explicitly ask annotators to consider the narrative schemas and to ignore the specific names of entities, only considering their roles. They do not define an exact measure of how distances between schemas are determined, nor do they instruct annotators to write down explicit schemas. Despite these limitations, they achieve comparatively good inter-annotator agreement (0.69 Krippendorff’s α) in narrative similarity of news articles.

3 Our Approach

Our model, called StoryEmb, is a causal language model whose last token representation is fine-tuned on similarity tasks using augmented data. Our model is trained to produce representations that are similar for multiple

summaries of the same story. As a foundation model, we use Mistral-7B (Jiang et al., 2023a). Specifically, we use E5 (Wang et al., 2024), an adapter-finetuned variant, trained using synthetic data, for similarity modeling. As story similarity is a complex task, we assume that a more capable model would perform better; due to hardware constraints, we chose a 7B parameter model.

We train our model using Gradient Cache (Gao et al., 2021) to enable large batch sizes on limited hardware while reaching identical results to traditional similarity training. In training, we optimize for reducing the cosine similarity between pairs of summaries labeled as the same while maximizing the cosine similarity between those pairs that, by nature of belonging to different works, are implicitly labeled as different. Our approach follows Gao et al. (2021) in using contrastive MSE-loss for similarity training. We use a batch size of 1000 positive pairs and in-batch negatives. For the optimizer, we use Adam with a learning rate of 5×10^{-5} and perform early stopping on a subset of pseudonymized summaries from the Tell-Me-Again dataset. The training is limited to the adapter parameters and, as we are training based on their weights, we follow Wang et al. (2024) and use LoRA with rank $r = 16$ and $\alpha = 32$. While our training setup differs in various details (we use a different loss and do not employ hard negatives), the training can be considered a continued fine-tuning of E5 with a similar objective, just focusing on narrative similarity.

Our training data is sampled from the Tell-Me-Again dataset but limited to only summaries with a minimum of 10 and a maximum of 50 sentences in length. This is motivated by the desire to exclude (a) very short synopses and loglines on the low end and (b) documents that are too memory-demanding on the high end. The length limit could be subject to further experimentation in the future. We evaluate whether the data augmentation approach – replacing names with alternative ones in a consistent manner – proposed by Hatzel and Biemann (2024) can improve the performance of a similarity model. To this end, we compare an augmented version of our model, trained on pseudonymized versions of the original summaries, and a non-augmented version, trained on the original summaries.

Following the E5 paper, we add a query prefix to each document. Through manual exploration on the development set, we selected the query, ‘Retrieve stories with a similar narrative to the given story:’. While many of the original applications of E5 follow an asymmetric setup where the query and the document are encoded using separate prompts, our prompt aligns well with one of their evaluation prompts: ‘Retrieve tweets that are semantically similar to the given tweet’.

BehnamGhader et al. (2024) have recently introduced a more sample-efficient way, called LLM2Vec, to train LLMs for sentence representations. In preliminary experiments, we found, perhaps in part due to length limitations in training as a result of the full-attention setup, an LLM2Vec-based model to perform inferiorly to our model.

4 Experiments

After training, we perform several downstream task experiments to explore the capabilities and characteristics of our narrative embeddings. Three experiments test narrative retrieval capabilities (Section 4.1). We also perform an experiment focused on narrative understanding (Section 4.2). All our experiments in this paper are limited to English data. Recall that our training data consists of pairs of story summaries automatically translated from various languages to English.

4.1 Narrative Retrieval

Using four different tasks, we test if our embeddings can be used for retrieving narratively similar stories. All retrieval tasks are performed using embedding cosine distances.

For the initial three retrieval experiments, those with gold data available, we follow Chaturvedi et al. (2018) in using P@1 (precision at one), in other words accuracy for the most relevant result. Additionally, we introduce the P@N (precision at n) metric to allow for easy interpretation of the results. It measures the precision in the N-top results, where [ILLEGIBLE] is the number of gold items in the respective cluster. For reference, we also include the more established information retrieval metrics of MAP (mean average precision), NDCG (normalized discounted cumulative gain), and R-Precision (Manning et al., 2008).

4.1.1 In-Task Performance

In prior work (Hatzel and Biemann, 2024), we tested various existing models on pseudonymized and non-pseudonymized versions of the dataset, finding that all models, especially smaller ones, perform very poorly on the pseudonymized versions. In the existing publication, we attribute this to those models’ reliance on entity names, showing that a bag-of-word system based only on entity mentions already performs well.

4.1.2 In-Domain Adaptation: Movie Remake Dataset

We expect retrieval performance on the movie remake task to be worse than on the Tell-Me-Again dataset, as one would expect summaries across remakes to show more variations than summaries sourced from various languages. This would align with our previous results (Hatzel and Biemann, 2024), where the best-performing model reached a P@1 of 64.4

4.1.3 Retellings

We collect a small set of summaries of works of fiction, each considered a retelling or a retelling’s original. The collection methodology amounted to prompting ChatGPT for close retellings to limit the variations in the narrative.¹ The model was essentially used to suggest retelling relationships, and the list was subsequently checked for validity using manual web searches. While we considered other approaches, such as using existing lists of retellings, we decided, in part due to a lack of authoritative lists of this nature, to retrieve very commonly mentioned pairs using a language model instead. After discarding various suggestions that did not have English Wikipedia articles with plot summaries, we ended up with 13 clusters of retellings totaling 30 story summaries.

Retellings often change the story in major ways, more so than we would expect in a movie remake. We expect retellings to deviate more from each other than both multiple summaries of the same story and summaries of movie remakes. However, they may retain similar or identical character names, a characteristic that is not aligned with our pseudonymized training data. Given these characteristics, we initially anticipated that our model would find the retelling retrieval task more challenging than identifying movie remakes. We release retelling the dataset, including the full summaries, alongside our code, in a format matching that by Chaturvedi et al. (2018) for easy comparison.

4.1.4 Segment Retrieval

To generalize these findings to a broader story retrieval problem, we perform an annotation-based experiment, asking LLM judges and human annotators to rate the narrative similarity of text pairs. While a human-curated dataset of similar story pairs may also be desirable, we do not see a clear path to creating one. A human judgment of similarity relies on recalling a large set of

¹See Appendix B for the prompt and further details

stories, which is not generally achievable with annotators. So, our experiment instead relies on testing pairs of texts that the model considers to be very similar or dissimilar using human annotators. We follow Chen et al. (2022a) in broadly annotating for similarity in narrative schemas without making them explicit during annotation. A more precise definition of narrative similarity on the basis of schemas could be the subject of future work, but we do not consider it essential for this limited-scale experiment.

For this experiment, we select a moderately sized fiction dataset in which we expect to find frequent occurrences of similar scenes. We select a set of public-domain detective novels for this purpose.²](<https://www.gutenberg.org/ebooks/bookshelf/30>) The novels are split into segments of no more than 2000 whitespace-separated tokens using a rule-based splitting solution.³](<https://github.com/umarbutler/semchunk>) Said segments are subsequently summarized using LLaMA3’s 70B⁴](<https://github.com/meta-llama/llama3>) (at full 16-bit precision) variant with the prompt “Please summarize the following text in three sentences or less.”. The resulting summaries are embedded using our StoryEmb model.

Initially, we remove all obviously similar pairs of summaries by discarding all pairs with a similarity higher than 0.3 according to MiniLM.⁵](<https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>) This ensures that duplicates that occur across documents in the dataset are not used as trivial examples of narrative similarity. We evaluate the similarity of the 50 most similar segment pairs and 50 least-similar pairs in two setups: (a) first with an LLM judge and (b) with a human judge. For the latter, we sample just 10

4.2 Narrative Understanding: ROCStories

Finally, we perform an experiment aimed at validating the common-sense understanding of our model using the Story Cloze Task. In story cloze, given a common-sense story of four sentences, the system has to select the final fifth sentence of the story from two choices: an incoherent but surface-level-consistent ending and the correct and semantically coherent one. To test our embeddings, we take an unconventional approach to inference on this task, enabling evaluation without a classification head or similar techniques. We embed three components: the first four sentences of the story that we refer to as the anchor a and two variants of the entire five-sentence story with either the second or the first option: s_1 and s_2 . Our system predicts

²](<https://www.gutenberg.org/ebooks/bookshelf/30>)

³](<https://github.com/umarbutler/semchunk>)

⁴](<https://github.com/meta-llama/llama3>)

⁵](<https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>)

the story that is closer to the anchor embedding. The intuition behind this is that a good story embedding already encodes expected outcomes, leading to a vastly different embedding for the incorrect, unfitting ending.

$$m(a, s_1, s_2) = \begin{cases} 1 & d(a, s_1) < d(a, s_2) \\ 2 & d(a, s_1) \geq d(a, s_2) \end{cases} \quad (1)$$

See Equation 1 for a more formal description, where d is an arbitrary distance measure, in our case cosine distance. For reference, we test not only our StoryEmb model but also other embedding models.

5 Results

As seen in Table 1, our StoryEmb model achieves state-of-the-art results on the Tell-Me-Again dataset, outperforming all other tested models in all but one metric. On the test set, our model, trained only using the augmented Wikipedia summaries, reaches a P@N of 65.89

We also test a pre-trained doc2vec model (Lau and Baldwin, 2016) as a more traditional baseline with no inherent length limitation. Outside of our own model’s performance, it is interesting to see doc2vec outperform E5 by far on the pseudonymized version of the dataset; the static-embedding model exhibits no noticeable drop in performance from the standard to the pseudonymized setting (in fact, the results on the pseudonymized version are marginally better for all metrics). Upon consideration, this is not surprising as the static word embeddings in doc2vec may have a hard time with generic entity names, especially personal names.

While the performance increase on the pseudonymized texts is expected, it is surprising that, even for the non-pseudonymized texts, the model trained on augmented data performs better. As noted earlier, our model’s training was stopped early based on the performance on the pseudonymized texts (for both model variants). The training finished after just three training steps (after seeing no improvements for two more steps), with each step taking roughly [ILLEGIBLE] min on two A100 GPUs. In fact, the unaugmented model continues to improve on the non-pseudonymized data afterward, presumably due to an ever-increasing focus on entity names as a shortcut to solving the task. [ILLEGIBLE]

[ILLEGIBLE]

Table 1: Retrieval performance on the Tell-Me-Again test set by Hatzel and Biemann (2024), with and without their anonymization strategy.

[ILLEGIBLE]

Table 2: Test set retrieval performance on the dataset by Chaturvedi et al. (2018), with and without the anonymization strategy by Hatzel and Biemann (2024) applied to the dataset. “+2 steps” denotes two additional steps of training.

5.1 Movie Remakes

The results for the movie remake dataset listed in Table 2 are state-of-the-art for said dataset with a top P@1 score of 83.26

We also provide results for the unaugmented StoryEmb model trained for two more steps, thereby almost doubling the fine-tuning data. Yet, despite the additional training data, our non-augmented model on the non-pseudonymized dataset does not meaningfully improve. Additional training only improves the P@1 score of 63.09

An interesting takeaway from the results on the movie remake dataset is a very pronounced drop in the performance of Sentence-T5 as compared to the Tell-Me-Again results. While the model showed a P@N of 94.98 on the non-pseudonymized Tell-Me-Again data, its performance dropped by more than 17 points to 77.61

5.2 Retellings

The retrieval performance on the retelling dataset tests our model’s capabilities in an alternative scenario with different requirements. On this dataset, Sentence-T5 outperforms our model by a considerable margin, reaching a P@1 of 70.0

To test this hypothesis, we add the summaries from the movie remake dataset as distractors to the task. In this setup, we see the margin by which T5 overperforms shrink considerably, especially for the P@N metric, where the best Sentence-T5 model now only outperforms our unaugmented model by roughly 2 points, with a score of 57.69

Interestingly, and despite the much smaller dataset size of only 30 rather than 266 summaries, our model’s and the baselines’ retrieval performance is much worse than on the movie remake dataset. This indicates that identifying

[ILLEGIBLE]

Table 3: Retrieval performance on retelling dataset introduced in Section 4.1.3, optionally with the movie remakes added as distractors.

[ILLEGIBLE]

Table 4: Mean narrative similarity score on a scale of 1–10 in top vs. bottom ranked scenes in terms of similarity as judged by an LLM judge or an annotator, after removing obvious duplicates. The first author performed the annotations.

retellings is a challenging task. At the same time, it is unclear if retellings are best identified using narrative features, given that they may only align with the story’s themes. Our augmented models’ underperformance may indicate that a name-focused approach is better suited to this task.

5.3 Scene Retrieval

Table 4 shows the narrative similarity ratings of our LLM judge and human annotator (the first author of this paper). The LLM judge favors the StoryEmb model when operating on the summaries of the retrieved segments, with the score increasing from 5.1 to 5.36 out of 10 when using our model instead of E5. This difference is much more pronounced when the LLM judge operates on the full segments instead. While our model is still rated at 5.36, the E5 model now only gets a score of 4.94. Our model also does better at retrieving dissimilar passages. We consider those passages retrieved by StoryEmb to have an average similarity of 3.6/10, in our annotations, whereas the segments retrieved by E5 score 4.6/10.

The LLM judgments on the full-text segments retrieved by StoryEmb are significantly ($p < 0.05$) more narratively similar than those retrieved by E5. We use the Mann-Whitney U significance test (Mann and Whitney, 1947), as a normal distribution cannot be safely assumed. Remember that these scores are achieved on a set of segments prefiltered to remove obviously similar examples.

5.4 Story Cloze: ROCStories

Table 5 lists the results of our model on the ROCStories dev set. An accuracy of almost 90

The results show that an expected event in the story changes the embedding less than an unexpected one. Thus, this experiment indicates a high level of narrative understanding exhibited by our story embeddings.

6 Approximate Attribution

To inspect which aspects our StoryEmb model focuses on, we apply an attribution approach for sentence encoders (Moeller et al., 2024). The approach builds on the idea of integrated gradients ([ILLEGIBLE]), extending the concept to Siamese networks, specifically sentence encoders. In essence, the approach samples gradients along an interpolation from a semantically neutral sequence to the analyzed sequence, identifying features that the output is sensitive to. The result is a token-token matrix across two input sequences, signifying the contribution of any two terms to the overall similarity of the two sequences.

Moeller et al. (2024) operate only on models that use average pooling across tokens. We adapt their implementation to work with decoder-only models and use 50 interpolation steps. E5 employs last-token pooling, using the last token’s hidden state as a sentence representation. As a result, the attribution scores are muddled: information needs to flow to the last token, leading to the majority of the similarity being explained by changes in the last token’s representation. In an effort to get interpretable attribution scores despite the pooling approach, we display the delta in attribution scores from the E5 model to our StoryEmb model instead.

Figure 1 illustrates that our augmented-data approach does, in fact, place less emphasis on named entities than the vanilla E5 model. We compare similarity across the two sentences ‘Alice wakes up.’ and ‘Alice falls down.’. Generally, negative values in an attribution indicate that two specific tokens interact to reduce the similarity of the two sentences. The attribution scores Figure 1 are computed by deducting the E5 attribution values from the StoryEmb attribution values. Thus, a negative value means our StoryEmb model places comparatively less emphasis on said value. In this case, our fine-tuned model places much less importance on the name ‘Alice’ than the original E5 model does.

To generalize from the single example, we collect attribution scores for 50 random sentences from the STS benchmark dataset’s test set (Cer et al., 2017). We collect average attribution scores for part-of-speech and named-entity tags, showing the results in Table 6. One can see the expected effect from our training on the part of speech tags; the model places much less

[ILLEGIBLE]

Figure 1: Attribution scores on individual tokens in the final layer of our StoryEmb model are shown as a delta from the E5 model. Negative scores indicate less contribution to the similarity in the StoryEmb model. In the example, it seems clear that less emphasis is placed on named entities.

[ILLEGIBLE]

Figure 2: [ILLEGIBLE]

emphasis on proper nouns (PROPN), while verbs (VERB) contribute slightly more to similarity scores. This effect is also visible for the named entity tags, with persons (PERS) and organizations (ORGS) being considered less relevant. Using the named-entity tags, however, we can also observe an unwanted side effect of the data augmentation; as dates are not removed by the data augmentation, they can still serve as a shortcut for solving the task, and our StoryEmb model prioritizes them.

We have restricted the analysis to single sentences as it is computationally expensive and could not feasibly be performed on entire stories.

7 Qualitative Exploration

In general, the manual evaluation of similarity has [ILLEGIBLE] issue that many datasets contain some form of [ILLEGIBLE], leading to the issue that trivially similar narratives are retrieved first. We explored similar segments, using the approach from Section 4.1.4, excluding trivially similar texts. We found many instances of meaningfully narratively similar texts being retrieved by StoryEmb. In Figure 2, we show a pair of segment summaries from the detective novel dataset, with the segment from ‘The Triumphs of Eugene Valmont’ being considered the 17th closest, by StoryEmb, to the ‘In the Fog’ segment. [ILLEGIBLE], meanwhile, only considers it to be the 221st closest segment. The two segments share a very similar narrative of a necklace theft with a subsequent arrest but lack any shared named entities.

8 Conclusion

In this work, we presented an approach to creating embeddings that represent stories, specifically their narratives. Our StoryEmb model, trained on the

Tell-Me-Again dataset, shows state-of-the-art performance on both the corresponding test split and on the movie remake dataset (Chaturvedi et al., 2018), far outperforming both recent LLM-based models and the static-embedding baseline. We demonstrate the model’s retrieval capability in practice on summaries of passages of literary texts. Further, we explore the retrieval of retellings on a small-scale dataset, opening up another potential application avenue. Our StoryEmb model also performs strongly on the StoryCloze task, indicating some level of narrative understanding.

On the movie remake dataset, we clearly demonstrate the effectiveness of the Tell-Me-Again data augmentation approach, producing better results on both pseudonymized and non-pseudonymized versions of the texts. Similarity score attribution indicates that the data augmentation techniques employed have the desired effect: making the model place less emphasis on names.

9 Future Work

In the future, we expect to explore application fields for story embeddings further. Outside of employing larger foundation models, the contrastive learning approach can potentially be improved; we assume that one may also, at the cost of compute resources, [ILLEGIBLE] a better model by not using a pre-existing similarity model but instead starting from a plain language model.

Similarly, there is potential to generate better training data by (a) improving the pseudonymized versions or (b) creating new summaries, potentially by combining multiple existing ones using LLM prompting.

10 Limitations

[MISSING] summaries. We do not expect this to be a substantial issue as we expect the different language versions to individually be trained on, as evidenced by relatively poor performance without further training. However, it is possible that the data augmentation strategy has limited success with entity names being inferred from the unredacted text seen in training.

It is possible that, given further hyperparameter tuning, the results could be noticeably improved. Contrastive learning, especially in the image space, has seen many optimizations. Due to resource constraints, this work was out of scope for this study. In initial experiments, we did not succeed with batch-sampling techniques, but it can be assumed that further exploration could yield improvements.

The representations we present lack interpretability compared to schema-based approaches to narrative modeling. At present, we cannot confidently identify which aspects of a story and its narrative are captured in our embeddings, and more work is required to understand exactly which information is captured by StoryEmb.

Our ROCStories results were obtained on the development set as the test set is privately held. We contacted the original authors and researchers who recently reported results on the dataset but were unable to get our predictions for the private test set scored. We have no reason to believe our performance on the test set would be worse.

11 Ethical Considerations

We do not see any major ethical problems. The data augmentation strategy in the original Tell-Me-Again dataset picks names based on US census statistics, potentially contributing to a system that may be regionally and culturally biased. This limitation is inherent to many NLP systems and should be addressed before approaches like this are used productively.

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References

- Parishad BehnamGhader, Vaibhav Adlakha, Marius Humanities2022CombinedAbstracts,Tokyo,Japan.
 Grant Glass. 2022. An Adaptive Methodology: Machine Learning and Literary Adaptation. It Digital
 Mosbach, Dzmitry Bahdanau, Nicolas Chapados, and
 DH2022 Local Organizing Committee.
 5940
 Melanie Goldman. 2023. The Rise of Fairytale Retellings in Publishing. Publishing Research Quarterly,39(3):219-233.
 Mark Granroth-Wilding and Stephen Clark. 2016. What Happens Next? Event Prediction Using a Compositional Neural Network Model. In Proceedings of the AAN Conference on Artificial Intelligence, volume 30, pages2727-2733, Phoenix, Arizona, USA.

Hans Ole Hatzel and Chris Biemann. 2023. Narrative cloze as a training objective: Towards modeling stories with corpus-derived narrative event chains. In Proceedings of the 5th Workshop on Narrative Understanding, pages 30-42, Toronto, Canada. Association for Computational Linguistics.

Hans Ole Hatzel and Chris Biemann. 2024. Tell-Me-Again: A Dataset of Multiple Narrations of the Same Story. In Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, pages 21617-21642, Miami, Florida, USA. Association for Computational Linguistics.

Wen-Pin Hsiao, Alexander Hanbo Li, Mikhail Y. Malinin, and Mark Gales. 2024. ACL 2024 Shared Task: Multi-domain and Multi-lingual Repetition Robustness Testing for Text-to-Image Generation Models. In Proceedings of the 9th Workshop on Representation Learning for NLP (RepL4NLP-2024), pages 318-334, Bangkok, Thailand. Association for Computational Linguistics.

Jing Jiang, Bin Wu, and Haizhou Li. 2023a. Mistral-7B. [ILLEGIBLE]

Xiaolan Jiang, Chen Fang, Tao Shen, Zhongqi Li, and Zhen Fu. 2023b. Integrating semantics into narrative schema induction. In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, pages 13733-13748, Singapore. Association for Computational Linguistics.

Tian Jin, Michael Carini, and Dan Boneh. 2024. gpl- [ILLEGIBLE]

Guoqing Lau and Timothy Baldwin. 2016. An Empirical Evaluation of doc2vec with Practical Insights into Document Embedding Generation. In Proceedings of the 1st Workshop on Representation Learning for NLP, pages 78-86, Berlin, Germany. Association for Computational Linguistics.

Kyu Young Lee and Ke Jung. 2020. Event-aware sentence representation for narrative event prediction. In Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), pages 8421-8430, Online. Association for Computational Linguistics.

[ILLEGIBLE] and Toma Mikolov. 2014. Distributed Representations of Sentences and Documents. In Proceedings of the 31st International Conference on Machine Learning, volume 32 of Proceedings of Machine Learning Research, pages 1188-1196, Beijing, China. PMLR.

H. B. Mann and D. R. Whitney. 1947. On a Test of Whether one of Two Random Variables is Stochastically Larger than the Other. The Annals of Mathematical Statistics, 18(1):50-60.

Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schutze. 2008. Introduction to Information Retrieval. Cambridge University Press, Cambridge, UK.

Todor Mihaylov and Anette Frank. 2019. Story Cloze Ending Selection Baselines and Data Examination. In Proceedings of the 2019 Conference

of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers), pages 1877-1881, Minneapolis, Minnesota. Association for Computational Linguistics.

Timo Moeller, Divyansh Agarwal, Christin Seifert, and Katja Filippova. 2024. Efficient and Robust Attribution for Sentence Encoders. In Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers), pages 6877-6892, Mexico City, Mexico. Association for Computational Linguistics.

Nafise Sadat Moosavi, Mohammad Taher Pilehvar, and Nigel Collier. 2023. Prompting with story series: An approach to natural language prompt engineering. In Findings of the Association for Computational Linguistics: EMNLP 2023, pages 1413-1424, Singapore. Association for Computational Linguistics.

Nasrin Mostafazadeh, Nathanael Chambers, Xiaodong He, Devi Parikh, Dhruv Batra, Lucy Vanderwende, Pushmeet Kohli, and James Allen. 2016. A Corpus and Evaluation Framework for Deeper Understanding of Commonsense Stories. In Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, pages 839-849, San Diego, California. Association for Computational Linguistics.

Jiacheng Ni, Gustavo Hernandez Abrego, Noah Constant, Matthew E. D. Lasko, Jannik Striitz, Muthukumar Venkatesan, and Tong Zhao. 2022. Sentence-T5: Scalable Sentence Encoders from Pre-trained Text-to-Text Models. In Findings of the Association for Computational Linguistics: ACL 2022, pages 1864-1874, Dublin, Ireland. Association for Computational Linguistics.

Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. In Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), pages 3982-3992, Hong Kong, China. Association for Computational Linguistics.

Jennifer Springer, Marie Niaudeau, Tian Bai, and Michael Hirsch. 2024. Repeat: Echoing Inputs to Improve Sentence Representations. In International Conference on Learning Representations, Online. OpenReviewnet.

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A LLM Judge

We use GPT4-o, specifically gpt-4o-2024-05-13, in a multi-turn setup for similarity evaluation. The model is first asked to describe similarities and differences in the narratives of both segments, and only in a second step is it asked to submit a rating. Our conversation template looks as follows:

System: ‘You are a helpful assistant specializing in fictional texts and their narrative.’

User: ‘In which respects, particularly focused on the narrative, are the following two stories similar and dissimilar? Focus on the structure of the story. Do not focus on specific names or places.’

Assistant: ... User: ‘Now based on your assessment, rate the similarity of the two stories on a scale of 1-10 where 10 is very similar. Please use the format [ILLEGIBLE]’ Assistant: ...

The sequence ‘...’ denotes that we let the LLM generate the response. We use a temperature of 0 for both generation steps.

B Retelling Dataset

We generated a retelling dataset by first prompting ChatGPT with pairs of retellings and originals and then selecting those pairs that contain English plot summaries on Wikipedia for both items. The prompt we used was:

‘Give me a list of five retellings of the fairytale Cinderella. Make sure that they follow the original plot closely, not just the themes.’
ILLEGIBLE

‘Give me a list of five retellings of a given story...’

This process was repeated for a variety of works. We then manually validated those retellings. While each retelling must have an English Wikipedia plot summary to be included, we also require the retelling to have an explicit mention of its original, for instance in Wikipedia or another web source. We tested 13 clusters totaling 30 story summaries.

C Retelling Dataset Results

In Table 7, we list the full results for the retellings experiment from Section 5.3. The table includes results with and without distractors. Note that, due to the small dataset size, the results should be interpreted with caution.

[ILLEGIBLE]

Table 5: Retrieval performance on retelling dataset introduced in Section 4.1.3, optionally with the movie remakes added as distractors.

D Data Code Availability

Our code and models are available online: <https://github.com/uhh-lt/story-emb>.